

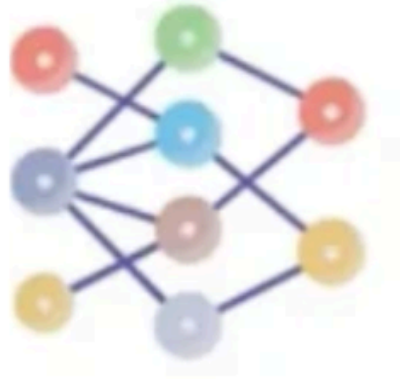
MEMBERSHIP INFERENCE ATTACK

presented by

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MOTIVATION

Was



trained on the example:



“ You have the right to remain in the training set.”

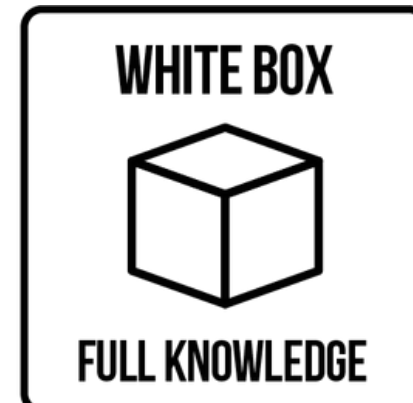
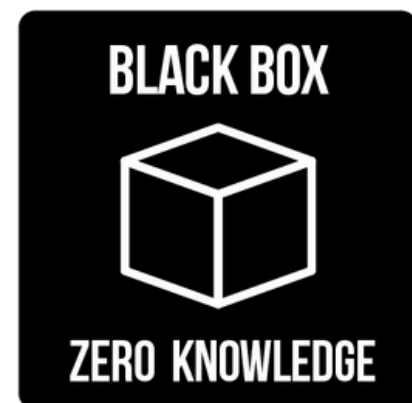
WHY?

- find out **what data** a company trained their models on
- **data extraction** from trained models
- **audit** a privacy algorithm
- develop **counter measures** to malicious actors
- specific **metrics** for privacy in ML

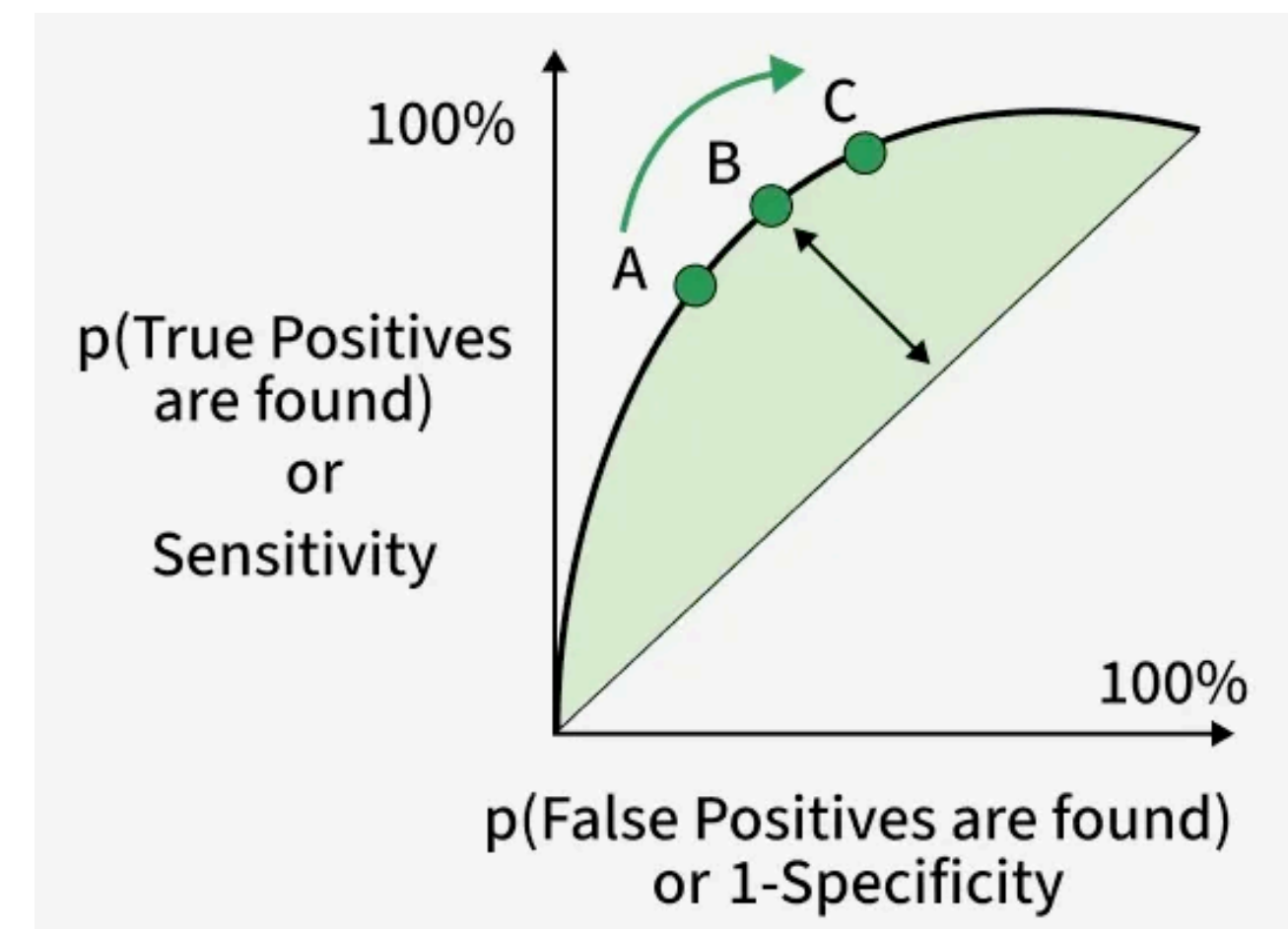


DEFINITIONS

GREY BOX SETTING



AUC-ROC & TPR@FPR=0.1

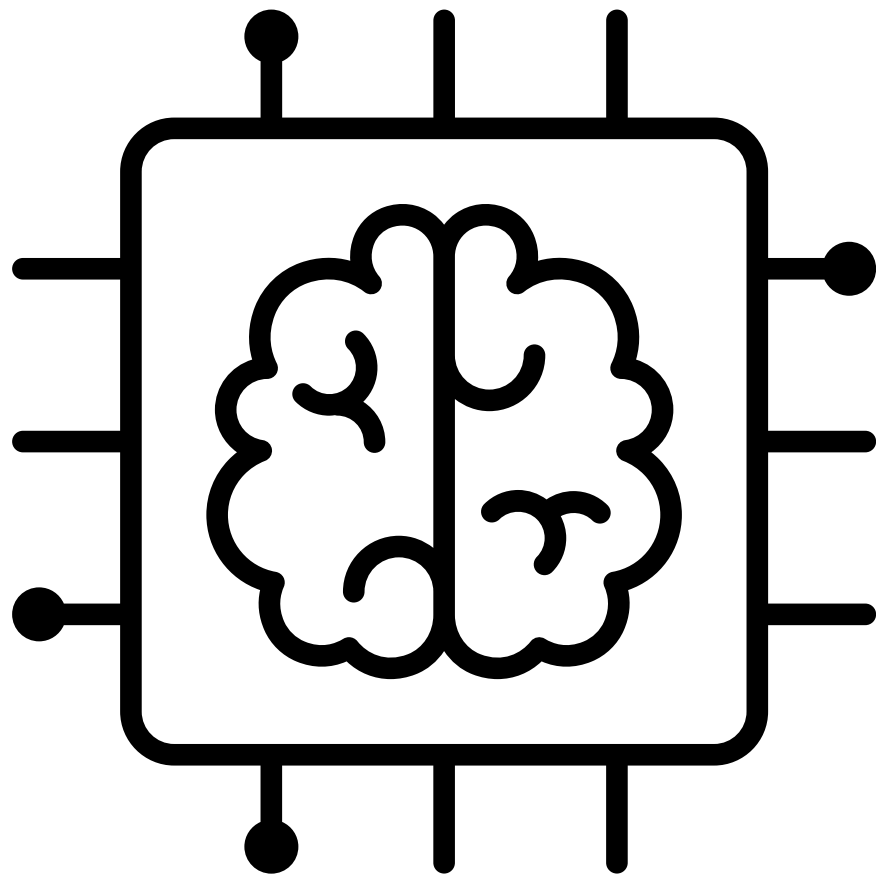


PREVIOUS WORK

“What Hard Tokens Reveal: Exploiting Low-confidence Tokens for Membership Inference Attacks against Large Language Models” by Jawad et al., 2026

“Membership Inference Attacks against Language Models via Neighbourhood Comparison” by Mattern et al., 2023

“Tracing Training Footprints: A Calibration Approach for Membership Inference Attacks Against Multimodal Large Language Models” by Zheng et al., 2025



LLM domain:

- Patient discharge notes
- Training: 50k
- Validation: 10k
- Test: 15k

EDA:

- Similar **text uniqueness** for members and non-members
- **Text lengths** evenly distributed across train/val/test
- **Image sizes** all the same

DATA

VLM domain:

- Image-caption pairs
- Training: 6k
- Validation: 12k
- Test: 6k
- Type: VLM has seen
 - Text and image
 - Image
 - Text
 - Neither



METHODS

TEXT-ONLY SETTING

SmolLM2-135M, SmolLM2-360M

Features: loss, perplexity, entropy, token probability statistics, Min-K%, HT-MIA, token-level LLR, cross-model delta/ratio

CLASSIFIERS

Logistic Regression, Random Forest, Gradient Boosting, XGBoost, weighted ensemble

(learn from features)

VLM SETTING

fine-tuned SmolVLM + base/reference SmolVLM (original pretrained SmolVLM)

Features: target-reference features, temperature sensitivity, Rényi-based features, image perturbation, neighbor comparison, PC-MMIA

(how confidence changes)

- Text-only results improved steadily Baseline Kaggle AUC: **0.89291**
- Best final Kaggle AUC: **0.95105**

hard-token features and target-reference comparisons are better at capturing signals

- VLM results were lower but improved Baseline Kaggle AUC: **0.76850**
- Best final Kaggle AUC: **0.83701**

multimodel membership inference attack is more complex, adding many features did not always help; careful feature selection mattered more

EXPERIMENTS

RESULTS

Phase	Milestone	Methods Used	Validaton AUC	Kaggle AUC
Text	1	Logistic Regression Baseline: 9 features	0.8852	0.89291
Text	2	Min-K%, Cross-Model Deltas, Partial SPV-MIA w/ XGBoost	0.8955	0.90549
Text	3	HT-MIA & LLR Features, Ensembling	0.9487	0.95105
VLM	4	XGBoost Baseline	0.7793	0.76850
VLM	5	Temperature, MaxRenyi-K% and ModRenyi, ICIMIA	0.7846	0.77626
VLM	6	Feature removal, PC-MMIA	0.8533	0.83701

CONCLUSION

- **Compute and time constraints were a big factor.** We had to skip or reduce the implementation of some of the strategies we read about, due to constraints on time and compute
- **Comparative Signals are Superior:** Across all milestones, models with features that compare fine-tuned model to reference (base) model outperformed models that only used absolute metrics like raw loss or perplexity.
- **Model Size Matters:** Larger LLMs produced cleaner memorization signals, confirming that increased capacity often leads to higher vulnerability to MIA.
- **Feature Selection Over Complexity:** In the final milestone, the "filtered feature approach" outperformed more complex ensembling, suggesting that adding too many features can introduce noise and "confuse" the learning signal.

OUR TEAM



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